

Agentic GraphRAG for Financial Question Answering

Corporate financial reports, particularly SEC 10-K filings, contain extensive textual and numerical information spread across multiple sections, making accurate financial question answering a challenging retrieval and reasoning task. Conventional Retrieval-Augmented Generation (RAG) systems primarily depend on vector-based similarity search over unstructured passages, which often proves insufficient for complex queries requiring reasoning across multiple entities, fiscal years, and interconnected financial concepts. Graph-based RAG (GraphRAG) overcomes this by organizing information as structured knowledge, while Agentic RAG further enhances retrieval by enabling a language model to plan, select, and coordinate retrieval strategies according to query needs. This project proposes an Agentic GraphRAG framework for financial question answering using the FinReflectKG knowledge graph built from SEC 10-K filings. The system will integrate graph traversal, hybrid document retrieval, and symbolic financial computation to retrieve supporting evidence, perform calculations such as growth rates and financial ratios, and generate responses traceable to underlying graph facts. The framework specifically targets improved temporal and cross-company reasoning, and introduces confidence-based adaptive routing, allowing simple questions to follow lightweight retrieval paths while reserving multi-tool reasoning for complex analytical queries. Performance will be evaluated through a staged ablation — no-retrieval, vector RAG, graph-enriched RAG, and agentic GraphRAG — using the FinReflectKG-MultiHop benchmark across temporal, cross-company, and intra-document reasoning tasks, with answer accuracy, retrieval efficiency, latency, and evidence traceability as primary metrics. By combining structured financial knowledge with autonomous reasoning and transparent evidence retrieval, the framework aims to develop a more accurate and explainable financial question-answering system.